

Legendre Polynomials in the Rvachev Up Dictionary

Symmetric finite synthesis, closed Fourier–Legendre loops,
and transmuted spectral geometry

Research report for the ProveIt Fabius–Rvachev project

29 August 2026

Abstract

Let $u = \text{up}$ denote Rvachev’s compactly supported C^∞ atomic function, normalized as an even probability density on $[-1, 1]$. The repository under study contains two ingredients that naturally point toward a closed loop: an exact finite representation of every polynomial by shifted copies of u , and the absolutely and uniformly convergent Fourier–Legendre expansion

$$u(x) = \sum_{n \geq 0} u_n P_{2n}(x), \quad -1 \leq x \leq 1.$$

If

$$Q_n = \mathcal{D}_u P_n, \quad \mathcal{D}_u = M_u(D)^{-1},$$

where M_u is the moment generating function of the up-law, then the required Legendre specialization is

$$P_{2n}(x) = \frac{1}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n).$$

Substitution gives an exact self-representation of u by finite cancellation blocks. The grouped order is essential: the repeated atom coefficients do not form an unconditional scalar series.

Beyond this specialization, the report develops several corpus-relative new results. The deconvolved Legendre polynomials are orthogonal for a positive pullback inner product and diagonalize the conjugated nonlocal operator

$$\mathcal{L}_u = -D(1 - X_u^2)D, \quad X_u = x - \frac{M'_u(D)}{M_u(D)}.$$

This gives an operator-valued three-term recurrence, a transmuted Christoffel–Darboux identity, exact continuous–discrete biorthogonality, a Gram-metric projector, and a positive two-sided factorization of the even Legendre projection kernel by up-atoms. An exact Favard obstruction explains why this orthogonality is not generated by an ordinary scalar weight.

A second new strand exploits reflection symmetry in the one-cell up-spline space. At even degree $2N$, its reflection-even part is exactly the centered even polynomial space. The $N + 1$ central reflected pairs

$$\psi_{-r}(t) + \psi_{r+1}(t), \quad r = 0, \dots, N,$$

therefore become polynomial automatically. Their independence is certified by an explicit odd determinant

$$\det B_N = (-1)^N (C_N \sigma_N - 1), \quad C_N = 2^{1+N(2N-1)},$$

where

$$\sigma_N = [z^N] \frac{(1 - z)^{2N-2}}{\prod_{m=1}^{2N-1} (1 - z^{2^m})}.$$

Consequently every P_{2m} with $m \leq N$ has a unique centrally symmetric common-scale representation using exactly $N + 1$ pair blocks. Substitution into the Legendre partial sum gives a new symmetric finite closed loop converging to u in every C^r norm. Exact symbolic checks, numerical experiments, conjectures, and a fully commented Python program are included.

Contents

1	Scope, corpus audit, and status discipline	4
1.1	Audit strategy	4
1.2	Meaning of “new”	4
2	The up-function, its probability law, and Appell calculus	5
2.1	Normalization and Fourier product	5
2.2	Probability representation and moment generating function	5
2.3	Reciprocal MGF, Bell polynomials, and deconvolution	6
2.4	Thue–Morse and q-integer products	7
3	Imported arbitrary-polynomial synthesis	7
3.1	Global common-scale formula	7
3.2	Sharp scale, local dimension, and endpoint basis	8
3.3	The one-dimensional defect	8
4	Explicit Legendre specialization	8
4.1	Deconvolved Legendre polynomials	8
4.2	Finite shift-only representation	9
4.3	Endpoint compression	9
5	The canonical Fourier–Legendre expansion and first closed loop	10
5.1	Exact coefficients	10
5.2	Termwise closure	10
5.3	One common lattice per partial sum	10
5.4	Why grouping is essential	11
6	A hidden orthogonal family: pullback Legendre geometry	11
6.1	Pullback inner product and probabilistic form	11
6.2	Transmuted Rodrigues formula and recurrence	12
6.3	Conjugated Sturm–Liouville operator	13
6.4	Transmuted Christoffel–Darboux kernel	13
7	Continuous–discrete biorthogonality and Gram geometry	14
7.1	A metric projector in atom coefficient space	15
8	Two-sided up-atom factorization of the even projection kernel	16
9	Parity anatomy of the local up-spline space	16
10	A central reflected-pair basis and its q-determinant	17
10.1	Central pairs	17
10.2	Endpoint-tail generating functions	17
10.3	The determinant certificate	18
10.4	The central basis theorem	20

10.5 Arithmetic and the binary-partition threshold	20
10.6 Worked formulas	21
11 The symmetric common-scale closed loop	21
12 Superalgebraic convergence and derivative loops	22
13 Exact and numerical experiments	23
13.1 Determinant and reconstruction certificates	23
13.2 Cancellation in exact central atomizations	24
13.3 Legendre convergence	24
13.4 An exact spectral transition at degree twelve	26
14 Minimal central Gram geometry and structural interpretation	27
14.1 Three exact coordinate systems	27
14.2 How the q-product enters	28
14.3 Why closure does not force analyticity	28
15 Conjectures and frontier problems	28
16 Conclusions	30
A Exact coefficient algorithms	30
A.1 Moments and reciprocal-MGF coefficients	30
A.2 Deconvolved Legendre coefficients	30
A.3 Endpoint reduction	31
A.4 Binary-partition coefficient	31
A.5 Central coefficients	31
B Reproducibility inventory	31

1 Scope, corpus audit, and status discipline

1.1 Audit strategy

The recursive docs tree of the ProveIt Fabius-function project was surveyed through its live manifest, directory structure, consolidated source maps, and provenance notes. The full text of the directly relevant current TeX volumes was then audited, with targeted searches across adjacent live volumes. Files marked by the repository as byte-identical, absorbed, superseded, or provenance-only were treated as historical witnesses rather than mathematically independent sources. This avoids counting editorial copies as separate prior results.

The principal nonduplication boundaries are summarized in Table 1. The repository’s primary Fabius–Rvachev exposition and polynomial-synthesis consolidation are cited as [5, 6]; the closest pre-existing loop drafts are cited as [7, 8].

Table 1: Repository sources most directly relevant to this report.

Source	Role here	Imported material or nonduplication boundary
<i>The Fabius Function and Rvachev’s Up-Function</i>	Primary exposition and formal-development crosswalk	Fourier and sinc products, MGF and moments, exact Legendre coefficients, absolute/uniform convergence, cosine series, Poisson identities
<i>Up Polynomial Synthesis</i>	Consolidated arbitrary-polynomial theorem	Scale classification, global/local synthesis, endpoint basis, local dimension $d + 2$, annihilator A_d , canonical defect, Appell–Bell–Bernoulli calculus
<i>Closing the Lagrange–Legendre–Rvachev Loop</i> (v3)	Closest existing loop report	Literal Legendre atomization, grouped closure, endpoint compression, deconvolved recurrence, factorial growth and nonflattening, one-sided kernel factorization
<i>Lagrange–Rvachev Closed-Loop Report</i> (v4)	Wider multiscale/Lagrange synthesis	Scale-ladder closure, node cancellation, two-atom Appell blocks, Lambert– W conditioning discussion
Adjacent frontier volumes	Cross-connections	Thue–Morse products, binary partitions, inverse/sampling identities, Lambert– W asymptotics, Bell and Bernoulli polynomial machinery

One recent loop draft already contains the literal formula for P_{2n} and its substitution into the Legendre expansion. This report therefore marks that operation as imported. Its new work begins with the pure Legendre transmutation, the parity-compressed local space, the central-pair determinant, the symmetric minimal-pair loop, the exact Gram geometry, and the two-sided kernel factorization.

1.2 Meaning of “new”

Throughout, “new” means *not found in the audited repository snapshot and not encountered in the targeted literature search used for this report*. It is not an unqualified claim of global priority. Imported statements are identified, proofs are supplied for new theorems, numerical observations are not promoted to theorems, and conjectures are explicitly labeled.

Contribution map. The report separates four levels of closure:

1. the imported literal shift-only Legendre atomization and grouped infinite loop;
 2. a new pullback-orthogonal, probabilistic, and nonlocal Sturm–Liouville interpretation of the deconvolved Legendre family;
 3. new exact Gram, biorthogonal, projector, and tensor-kernel identities at every finite cutoff;
 4. a new centrally symmetric $N + 1$ -pair basis with an odd q -product determinant, yielding a common-scale loop convergent in C^∞ .
-

2 The up-function, its probability law, and Appell calculus

2.1 Normalization and Fourier product

Let $u = \text{up}$ be the even C^∞ function supported on $[-1, 1]$, normalized by

$$u(0) = 1, \quad \int_{-1}^1 u(x) \, dx = 1.$$

With the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} \, dx,$$

its transform is

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi} \left(\frac{\xi}{2^j} \right), \quad \text{sinc}_{\pi}(t) = \frac{\sin(\pi t)}{\pi t}. \quad (2.1)$$

At a nonzero integer m ,

$$\text{ord}_{\xi=m} \widehat{u}(\xi) = 1 + \nu_2(m), \quad (2.2)$$

where ν_2 is the dyadic valuation. This zero multiplicity is the source of the sharp degree- d reproduction scale 2^d .

On $[0, 1]$, the Fabius function is related by

$$F(t) = u(t - 1), \quad (2.3)$$

with the standard continuation supplied by its functional equations. Dyadic values of F therefore encode exact moments and Legendre coefficients of u . The original nowhere-analytic C^∞ example goes back to Fabius [1].

2.2 Probability representation and moment generating function

Let $V_1, V_2, \dots$ be independent uniform random variables on $[-1, 1]$ and put

$$Z = \sum_{j=1}^{\infty} 2^{-j} V_j. \quad (2.4)$$

Then Z has density u . Its moment generating function is

$$M_u(z) = \mathbb{E} e^{zZ} = \prod_{j=1}^{\infty} \frac{\sinh(z/2^j)}{z/2^j} = \sum_{r=0}^{\infty} \mu_r \frac{z^r}{r!}. \quad (2.5)$$

The law is symmetric, so odd moments vanish. The first even moments are

$$\mu_0 = 1, \quad \mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}. \quad (2.6)$$

Writing

$$\log M_u(z) = \sum_{r \geq 1} \kappa_r \frac{z^r}{r!},$$

one obtains

$$\kappa_{2r} = \frac{B_{2r}}{2r(1 - 4^{-r})}, \quad \kappa_{2r+1} = 0. \quad (2.7)$$

Thus the up-law is tied directly to Bernoulli numbers.

2.3 Reciprocal MGF, Bell polynomials, and deconvolution

Define rational coefficients γ_r by

$$\frac{1}{M_u(z)} = \sum_{r=0}^{\infty} \gamma_r \frac{z^r}{r!}. \quad (2.8)$$

In terms of complete Bell polynomials,

$$\gamma_r = Y_r(-\kappa_1, \dots, -\kappa_r). \quad (2.9)$$

The first nonzero values are

$$\gamma_2 = -\frac{1}{9}, \quad \gamma_4 = \frac{31}{675}, \quad \gamma_6 = -\frac{2347}{59535}, \quad \gamma_8 = \frac{1904369}{32531625}. \quad (2.10)$$

The finite-on-polynomials deconvolution operator is

$$\mathcal{D}_u = M_u(D)^{-1} = \sum_{r \geq 0} \gamma_r \frac{D^r}{r!}, \quad (2.11)$$

with inverse

$$M_u(D) = \sum_{r \geq 0} \mu_r \frac{D^r}{r!}. \quad (2.12)$$

The monomial images

$$A_n(x) = \mathcal{D}_u x^n = \sum_{r=0}^n \binom{n}{r} \gamma_r x^{n-r} \quad (2.13)$$

form an Appell family with generating function

$$\sum_{n \geq 0} A_n(x) \frac{z^n}{n!} = \frac{e^{xz}}{M_u(z)}. \quad (2.14)$$

The probability representation supplies a useful interpretation. For every polynomial f ,

$$(M_u(D)f)(x) = \mathbb{E}f(x + Z). \quad (2.15)$$

Hence $q = \mathcal{D}_u p$ is characterized by the exact deconvolution equation

$$\mathbb{E}q(x + Z) = p(x). \quad (2.16)$$

2.4 Thue–Morse and q -integer products

Let $\varepsilon_j = (-1)^{\text{wt}(j)}$ be the Thue–Morse signs. The finite Prouhet product is

$$\Theta_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}) = \sum_{j=0}^{2^m-1} \varepsilon_j z^j. \quad (2.17)$$

The annihilator governing local up-atom dependencies is

$$\mathcal{A}_d(z) = \prod_{m=1}^d [2^m]_z = \prod_{m=1}^d \frac{1 - z^{2^m}}{1 - z} = \prod_{\ell=1}^d (1 + z^{2^{\ell-1}})^{d-\ell+1}. \quad (2.18)$$

It is monic and palindromic, with

$$R_d := \deg \mathcal{A}_d = 2^{d+1} - d - 2, \quad \mathcal{A}_d(1) = 2^{d(d+1)/2}. \quad (2.19)$$

The determinant in [Section 10](#) will involve a reciprocal of this q -product and therefore restricted binary partitions.

3 Imported arbitrary-polynomial synthesis

3.1 Global common-scale formula

Let $p \in \Pi_d$, where Π_d denotes the real polynomials of degree at most d , and set

$$M = 2^d, \quad q = \mathcal{D}_u p. \quad (3.1)$$

The physical-lattice form of the repository's synthesis theorem is as follows [6].

Theorem 3.1 (Imported exact polynomial synthesis). *For every $p \in \Pi_d$,*

$$p(x) = \frac{1}{M} \sum_{k \in \mathbb{Z}} q(k/M) u(x - k/M), \quad x \in \mathbb{R}. \quad (3.2)$$

The sum is locally finite. On $[-1, 1]$ it reduces to

$$p(x) = \frac{1}{M} \sum_{|k| < 2M} q(k/M) u(x - k/M). \quad (3.3)$$

Scaling check. The normalized-cell formula in the repository is

$$P(t) = \frac{1}{M} \sum_{k \in \mathbb{Z}} Q(k) u\left(\frac{t-k}{M}\right), \quad Q = M_u(MD)^{-1}P.$$

Apply it to $P(t) = p(t/M)$ and put $t = Mx$. Since $D_t = M^{-1}D_x$, $Q(k) = \mathcal{D}_u p(k/M)$, which gives (3.2). In the Fourier proof, nonzero aliases contain derivatives through degree d of $\hat{u}(M\xi)$ at integer frequencies; (2.2) makes them vanish. Compact support gives local finiteness. On $[-1, 1]$, centers with $|k| \geq 2M$ meet only the exterior or a flat support endpoint. $\square$

3.2 Sharp scale, local dimension, and endpoint basis

For the normalized one-cell atoms

$$\psi_k(t) = \frac{1}{M} u\left(\frac{t-k}{M}\right), \quad 0 \leq t \leq 1, \quad (3.4)$$

only the $2M$ centers $-M+1 \leq k \leq M$ are active. The repository proves that the integer-translate system at ratio $\rho \in \mathbb{N}$ has exact polynomial reproduction order $\nu_2(\rho)$. Thus $M = 2^d$ is the smallest integer ratio reproducing all of Π_d .

Set

$$V_d = \text{span}\{\psi_k|_{[0,1]} : -M+1 \leq k \leq M\}. \quad (3.5)$$

Then

$$\dim V_d = d+2, \quad V_d \cap \mathbb{R}[t] = \Pi_d. \quad (3.6)$$

The $d+2$ rightmost atoms

$$\mathcal{B}_d = \{\psi_k|_{[0,1]} : M-d-1 \leq k \leq M\} \quad (3.7)$$

form a basis of V_d . Any degree- d polynomial can therefore be compressed from exponentially many literal lattice atoms to $d+2$ atoms, although the resulting signed coefficients may be very large.

3.3 The one-dimensional defect

The quotient V_d/Π_d is one-dimensional. A canonical representative is

$$\mathcal{D}_d(t) = t^{d+1} + \mathcal{E}_d(t), \quad (3.8)$$

where

$$\mathcal{E}_d(t) = -\frac{(d+1)!}{(2\pi i)^{d+1}} \sum_{\substack{n \in \mathbb{Z} \\ n \text{ odd}}} \frac{\widehat{u}(n/2)}{n^{d+1}} e^{2\pi i n t}. \quad (3.9)$$

Thus

$$V_d = \Pi_d \oplus \text{span}\{\mathcal{D}_d\}. \quad (3.10)$$

For odd d , $\mathcal{E}_d$ is a cosine series; for even d , it is a sine series. This parity fact drives the central-pair construction.

4 Explicit Legendre specialization

4.1 Deconvolved Legendre polynomials

Let P_n be normalized by $P_n(1) = 1$ and define

$$Q_n(x) = \mathcal{D}_u P_n(x). \quad (4.1)$$

Because $\mathcal{D}_u$ is even, Q_n has the same parity as P_n . The standard coefficient formula

$$P_{2n}(x) = \sum_{j=0}^n a_{n,j} x^{2n-2j}, \quad a_{n,j} = \frac{(-1)^j}{2^{2n}} \frac{(4n-2j)!}{j!(2n-j)!(2n-2j)!} \quad (4.2)$$

gives the explicit Bell–Bernoulli form

$$Q_{2n}(x) = \sum_{j=0}^n a_{n,j} \sum_{r=0}^{n-j} \binom{2n-2j}{2r} \gamma_{2r} x^{2n-2j-2r}. \quad (4.3)$$

All coefficients are rational. The first examples are

$$\begin{aligned} Q_0(x) &= 1, \\ Q_2(x) &= \frac{9x^2 - 4}{6}, \end{aligned} \tag{4.4}$$

$$\begin{aligned} Q_4(x) &= \frac{4725x^4 - 7200x^2 + 1072}{1080}, \\ Q_6(x) &= \frac{654885x^6 - 1984500x^4 + 1344168x^2 - 114080}{45360}. \end{aligned} \tag{4.5}$$

The probabilistic characterization (2.16) becomes

$$\mathbb{E}Q_n(x + Z) = P_n(x). \tag{4.6}$$

4.2 Finite shift-only representation

Theorem 4.1 (Literal Legendre-to-up transform; imported specialization). *Let $M_n = 4^n$. For every $n \geq 0$ and $-1 \leq x \leq 1$,*

$$P_{2n}(x) = \frac{1}{M_n} \sum_{|k| < 2M_n} Q_{2n}(k/M_n) u(x - k/M_n). \tag{4.7}$$

Equivalently,

$$P_{2n}(x) = \frac{1}{M_n} \left[Q_{2n}(0)u(x) + \sum_{k=1}^{2M_n-1} Q_{2n}(k/M_n) (u(x - k/M_n) + u(x + k/M_n)) \right]. \tag{4.8}$$

Proof. Apply Theorem 3.1 with degree $d = 2n$, so $M = 2^{2n} = 4^n$, and $q = Q_{2n}$. Pair k and $-k$ using the evenness of both Q_{2n} and u . $\square$

Example 4.2 (A completely explicit formula for P_2). Since $M_1 = 4$ and (4.4) holds,

$$P_2(x) = \frac{1}{384} \left[-64u(x) + \sum_{k=1}^7 (9k^2 - 64) (u(x - k/4) + u(x + k/4)) \right]. \tag{4.9}$$

This is a finite linear combination of literal translates of the undilated up-function.

The odd family has the analogous antisymmetric formula. With $M = 2^{2n+1}$,

$$P_{2n+1}(x) = \frac{1}{M} \sum_{k=1}^{2M-1} Q_{2n+1}(k/M) (u(x - k/M) - u(x + k/M)). \tag{4.10}$$

4.3 Endpoint compression

Mapping $t = (x + 1)/2$ and applying the endpoint basis at degree $2n$ gives unique rational coefficients $b_{n,k}$, $M_n - 2n - 1 \leq k \leq M_n$, such that

$$P_{2n}(x) = \frac{1}{M_n} \sum_{k=M_n-2n-1}^{M_n} b_{n,k} u\left(\frac{x+1-2k}{2M_n}\right). \tag{4.11}$$

This uses $2n + 2$ atoms rather than $4M_n - 1$ literal shifts, but its centers cluster near one endpoint. The central construction in Section 10 will retain the same raw atom count while enforcing reflection symmetry.

5 The canonical Fourier–Legendre expansion and first closed loop

5.1 Exact coefficients

Evenness gives

$$u(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x), \quad u_n = \frac{4n+1}{2} \int_{-1}^1 u(x) P_{2n}(x) dx. \quad (5.1)$$

The repository proves absolute and uniform convergence and supplies the exact Fabius formula [5]

$$u_n = 4^{-n}(4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \times (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (5.2)$$

The first values are

$$\begin{aligned} u_0 &= \frac{1}{2}, & u_1 &= -\frac{5}{6}, & u_2 &= \frac{11}{30}, & u_3 &= \frac{143}{5670}, \\ u_4 &= -\frac{7291}{85050}, & u_5 &= \frac{99137}{2485890}, & u_6 &= -\frac{5726162257}{213774110550}. \end{aligned} \quad (5.3)$$

5.2 Termwise closure

Substituting (4.7) into (5.1) gives the requested loop.

Theorem 5.1 (Blockwise Legendre–Rvachev closed loop; imported). *For $-1 \leq x \leq 1$,*

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u(x - k/4^n). \quad (5.4)$$

The outer series converges absolutely and uniformly when each finite inner sum is retained as one block.

Proof. The inner sum, including 4^{-n} , is exactly $P_{2n}(x)$. Hence the n th block is $u_n P_{2n}(x)$. Since $\|P_m\|_{\infty} \leq 1$ and $\sum_n |u_n| < \infty$, the Weierstrass test applies. $\square$

The parity-compressed form is

$$u(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \left[Q_{2n}(0) u(x) + \sum_{k=1}^{2 \cdot 4^n - 1} Q_{2n}(k/4^n) (u(x - k/4^n) + u(x + k/4^n)) \right]. \quad (5.5)$$

5.3 One common lattice per partial sum

Let

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x), \quad C_N(x) = \mathcal{D}_u S_N(x) = \sum_{n=0}^N u_n Q_{2n}(x). \quad (5.6)$$

Since S_N has degree $2N$, the common lattice $M = 4^N$ gives

$$S_N(x) = \frac{1}{4^N} \sum_{|k| < 2 \cdot 4^N} C_N(k/4^N) u(x - k/4^N). \quad (5.7)$$

Every coefficient is rational. Endpoint compression gives the same partial sum with $2N + 2$ atoms.

5.4 Why grouping is essential

The coefficient of the repeated central atom $u(x)$ in the n th literal block is

$$\beta_n = \frac{u_n Q_{2n}(0)}{4^n}. \quad (5.8)$$

The audited loop report proves

$$\limsup_{n \rightarrow \infty} |\beta_n| = \infty, \quad (5.9)$$

using the factorial asymptotic

$$Q_{2n}(0) = \frac{2(-1)^n}{C} \frac{(4n)!}{(2n)!(4\pi)^{2n}} \left(1 + \frac{2\pi^2}{4n-1} + \mathcal{O}(n^{-2}) \right), \quad (5.10)$$

where

$$C = M_u(\pi i) = \prod_{j \geq 1} \frac{\sin(\pi/2^j)}{\pi/2^j}.$$

Thus (5.4) is a convergent series of exact cancellation blocks, not an unconditionally convergent series indexed by individual atoms.

6 A hidden orthogonal family: pullback Legendre geometry

The operator $\mathcal{D}_u$ becomes rapidly ill-conditioned with degree, but on every fixed polynomial space it is triangular with unit diagonal and hence invertible. This permits a finite-dimensional Hilbert geometry in which the Q_n are genuinely orthogonal.

6.1 Pullback inner product and probabilistic form

Put

$$T = M_u(D), \quad A = T^{-1} = \mathcal{D}_u. \quad (6.1)$$

For polynomials define

$$\langle f, g \rangle_u = \int_{-1}^1 (Tf)(x)(Tg)(x) dx. \quad (6.2)$$

Since T is injective on polynomials, this is positive definite. If Z_1, Z_2 are independent copies of (2.4), then

$$\langle f, g \rangle_u = \int_{-1}^1 \mathbb{E}f(x + Z_1) \mathbb{E}g(x + Z_2) dx. \quad (6.3)$$

Theorem 6.1 (Pullback orthogonality; new). *The deconvolved Legendre polynomials satisfy*

$$\boxed{\langle Q_m, Q_n \rangle_u = h_n \delta_{mn}, \quad h_n = \frac{2}{2n+1}.} \quad (6.4)$$

Consequently $\{Q_0, \dots, Q_d\}$ is an orthogonal basis of Π_d for $\langle \cdot, \cdot \rangle_u$.

Proof. By definition $TQ_n = P_n$, so

$$\langle Q_m, Q_n \rangle_u = \int_{-1}^1 P_m P_n = \frac{2}{2n+1} \delta_{mn}.$$

Triangularity of A preserves degree and leading coefficient, hence the basis claim. $\square$

The inner product is explicitly derivative-coupled:

$$Tf = f + \frac{1}{18}f'' + \frac{19}{16200}f^{(4)} + \frac{583}{42865200}f^{(6)} + \dots, \quad (6.5)$$

with termination on each polynomial.

Proposition 6.2 (No scalar-weight realization; new). *The pullback inner product cannot be represented in the form*

$$\langle f, g \rangle_u = \int f(x)g(x) d\rho(x)$$

for any finite signed measure ρ having all polynomial moments.

Proof. Multiplication by x is symmetric for every scalar moment form. Here $Tx = x$ and $T(x^2) = x^2 + \mu_2 = x^2 + 1/9$, so

$$\langle x, x \rangle_u = \int_{-1}^1 x^2 dx = \frac{2}{3}, \quad \langle 1, x^2 \rangle_u = \int_{-1}^1 \left(x^2 + \frac{1}{9}\right) dx = \frac{8}{9}.$$

The two quantities differ, proving the claim. $\square$

There is also an exact Favard obstruction. Let $\widehat{Q}_n$ be the monic normalization of Q_n . Direct calculation gives

$$\begin{aligned} \widehat{Q}_2 &= x^2 - \frac{4}{9}, & \widehat{Q}_3 &= x^3 - \frac{14}{15}x, \\ \widehat{Q}_4 &= x^4 - \frac{32}{21}x^2 + \frac{1072}{4725}. \end{aligned} \quad (6.6)$$

Hence

$$\boxed{\widehat{Q}_4 = x\widehat{Q}_3 - \frac{62}{105}\widehat{Q}_2 - \frac{8}{225}\widehat{Q}_0.} \quad (6.7)$$

A monic parity-alternating scalar orthogonal sequence would satisfy $\widehat{Q}_{n+1} = x\widehat{Q}_n - \lambda_n\widehat{Q}_{n-1}$ by Favard's theorem [2]. The nonzero $\widehat{Q}_0$ term in (6.7) rules this out for the entire Q_n family, even for an indefinite quasi-definite scalar functional. The correct orthogonality is operator-valued, not weighted-scalar.

6.2 Transmuted Rodrigues formula and recurrence

Since A commutes with D , Rodrigues' formula becomes

$$Q_n(x) = \frac{1}{2^n n!} D^n [A(x^2 - 1)^n]. \quad (6.8)$$

The bracket means that the differential operator $A = M_u(D)^{-1}$ acts on the polynomial seed.

To transport multiplication by x , use $f(D)x = xf(D) + f'(D)$. Define

$$X_u = AxT = x - \frac{M'_u(D)}{M_u(D)} = x + \Lambda(D), \quad (6.9)$$

where

$$\begin{aligned} \Lambda(z) &= -\frac{M'_u(z)}{M_u(z)} = -\sum_{r \geq 1} \kappa_{2r} \frac{z^{2r-1}}{(2r-1)!} \\ &= -\frac{z}{9} + \frac{z^3}{675} - \frac{2z^5}{59535} + \dots \end{aligned} \quad (6.10)$$

Conjugating the classical recurrence gives

$$(n+1)Q_{n+1} = (2n+1)X_u Q_n - nQ_{n-1}. \quad (6.11)$$

This is a three-term recurrence in the self-adjoint operator X_u , not in scalar multiplication by x . The extra lower terms in (6.7) are exactly what appears when X_u is replaced by x .

6.3 Conjugated Sturm–Liouville operator

Let

$$\mathcal{L} = -D(1-x^2)D, \quad (6.12)$$

so $\mathcal{L}P_n = n(n+1)P_n$.

Theorem 6.3 (Nonlocal Sturm–Liouville transmutation; new). *Define*

$$\mathcal{L}_u = A\mathcal{L}T. \quad (6.13)$$

Then

$$\mathcal{L}_u = -D(1-X_u^2)D, \quad \mathcal{L}_u Q_n = n(n+1)Q_n. \quad (6.14)$$

Moreover,

$$\langle f, \mathcal{L}_u g \rangle_u = \langle \mathcal{L}_u f, g \rangle_u \quad (f, g \in \mathbb{R}[x]). \quad (6.15)$$

Although $\mathcal{L}_u$ is formally of infinite differential order, it is locally finite on every polynomial.

Proof. Because A, T , and D commute and $TA = I$,

$$Ax^2T = Ax(TA)xT = (AxT)^2 = X_u^2.$$

Conjugating (6.12) proves the first formula. Next,

$$\mathcal{L}_u Q_n = A\mathcal{L}T(AP_n) = A\mathcal{L}P_n = n(n+1)Q_n.$$

Finally $T\mathcal{L}_u = \mathcal{L}T$, so classical integration by parts gives

$$\langle f, \mathcal{L}_u g \rangle_u = \int (Tf)\mathcal{L}(Tg) = \int \mathcal{L}(Tf)(Tg) = \langle \mathcal{L}_u f, g \rangle_u.$$

The factor $1-x^2$ removes the boundary term. □

Remark 6.4. For scalar-weight orthogonal polynomials, multiplication by x is represented by a self-adjoint Jacobi matrix and all zeros are real and simple. Here the self-adjoint coordinate is X_u , while x fails symmetry by Proposition 6.2. Complex zeros of sufficiently high Q_n are therefore compatible with the exact pullback orthogonality.

6.4 Transmuted Christoffel–Darboux kernel

Let

$$K_d(x, y) = \sum_{n=0}^d \frac{P_n(x)P_n(y)}{h_n} \quad (6.16)$$

be the classical Legendre projection kernel, and define

$$\tilde{K}_d(x, y) = \sum_{n=0}^d \frac{Q_n(x)Q_n(y)}{h_n} = A_x A_y K_d(x, y). \quad (6.17)$$

Proposition 6.5 (Pullback reproducing kernel; new). *For every $f \in \Pi_d$,*

$$f(x) = \left\langle f, \tilde{K}_d(x, \cdot) \right\rangle_u, \quad (6.18)$$

and

$$T_x T_y \tilde{K}_d(x, y) = K_d(x, y). \quad (6.19)$$

Proof. Expand $f = \sum_{n \leq d} a_n Q_n$ and apply [Theorem 6.1](#). The second identity follows from $TQ_n = P_n$. $\square$

The classical Christoffel–Darboux formula survives as an operator identity.

Theorem 6.6 (Operator Christoffel–Darboux identity; new). *Let $X_{u,x}$ and $X_{u,y}$ act in the indicated variables. Then*

$$(X_{u,x} - X_{u,y}) \tilde{K}_d(x, y) = \frac{d+1}{2} [Q_{d+1}(x)Q_d(y) - Q_d(x)Q_{d+1}(y)]. \quad (6.20)$$

Proof. The classical identity is

$$(x - y)K_d(x, y) = \frac{d+1}{2} [P_{d+1}(x)P_d(y) - P_d(x)P_{d+1}(y)].$$

Apply $A_x A_y$. Since $X_{u,x} A_x A_y = A_x x A_y$ and similarly in y , the left side becomes $(X_{u,x} - X_{u,y}) \tilde{K}_d$; the right side becomes the displayed Q -expression. $\square$

The reciprocal diagonal

$$\lambda_d^{(u)}(x) = \tilde{K}_d(x, x)^{-1} \quad (6.21)$$

is a transmuted Christoffel function. Its asymptotics are proposed as an open problem in [Section 15](#).

7 Continuous–discrete biorthogonality and Gram geometry

Fix a cutoff N and use the common physical lattice

$$M = 4^N, \quad \mathcal{K}_M = \{k \in \mathbb{Z} : |k| < 2M\}. \quad (7.1)$$

Put

$$\phi_k(x) = u(x - k/M), \quad D_{kn} = \frac{1}{M} Q_{2n}(k/M), \quad 0 \leq n \leq N, \quad (7.2)$$

and

$$G_{k\ell} = \int_{-1}^1 \phi_k(x) \phi_\ell(x) dx, \quad H = \text{diag}(h_0, h_2, \dots, h_{2N}). \quad (7.3)$$

The literal synthesis theorem is

$$\Phi(x)D = (P_0(x), P_2(x), \dots, P_{2N}(x)), \quad (7.4)$$

where $\Phi(x)$ is the row vector of atoms $\phi_k(x)$.

Theorem 7.1 (Exact Gram isometry; new). *The synthesis matrix satisfies*

$$D^\top G D = H. \quad (7.5)$$

Consequently, for every $c \in \mathbb{R}^{N+1}$,

$$\left\| \sum_{n=0}^N c_n P_{2n} \right\|_{L^2[-1,1]}^2 = c^\top H c = (Dc)^\top G (Dc). \quad (7.6)$$

Proof. The (m, n) entry of $D^\top G D$ is

$$\int_{-1}^1 \left(\sum_k D_{km} \phi_k \right) \left(\sum_\ell D_{\ell n} \phi_\ell \right) dx = \int_{-1}^1 P_{2m} P_{2n} dx = h_{2n} \delta_{mn}.$$

□

Define the continuous analysis matrix

$$B_{jk} = \frac{1}{h_{2j}} \int_{-1}^1 P_{2j}(x) \phi_k(x) dx, \quad j \geq 0. \quad (7.7)$$

Theorem 7.2 (Atom–Legendre biorthogonality; new). *For every $j \geq 0$ and $0 \leq n \leq N$,*

$$\boxed{\frac{1}{M h_{2j}} \sum_{k \in \mathcal{K}_M} Q_{2n}(k/M) \int_{-1}^1 P_{2j}(x) u(x - k/M) dx = \delta_{jn}.} \quad (7.8)$$

Equivalently, $BD = I$ on the first $N + 1$ columns.

Proof. Insert the synthesis of P_{2n} into the defining integral:

$$(BD)_{jn} = \frac{1}{h_{2j}} \int_{-1}^1 P_{2j} P_{2n} = \delta_{jn}.$$

□

7.1 A metric projector in atom coefficient space

Define

$$\Pi_{\text{cof}} = DH^{-1}D^\top G. \quad (7.9)$$

Corollary 7.3 (Gram-orthogonal coefficient projector; new). *One has*

$$\Pi_{\text{cof}}^2 = \Pi_{\text{cof}}, \quad G\Pi_{\text{cof}} = \Pi_{\text{cof}}^\top G, \quad (7.10)$$

with range $\text{col}(D)$ and trace $N + 1$.

Proof. Use (7.5). Cyclicity gives

$$\text{tr}(DH^{-1}D^\top G) = \text{tr}(H^{-1}D^\top G D) = N + 1.$$

□

The literal dictionary is redundant, so G is positive semidefinite rather than positive definite. All identities are nevertheless exact on $\text{col}(D)$. Restriction to an independent local basis—for example the endpoint basis or the central pair basis below—produces a positive definite Gram matrix.

A scalar trace identity follows from $BD = I$:

$$\frac{1}{M} \sum_{k \in \mathcal{K}_M} \int_{-1}^1 u(x - k/M) \left[\sum_{n=0}^N \frac{Q_{2n}(k/M) P_{2n}(x)}{h_{2n}} \right] dx = N + 1. \quad (7.11)$$

Remark 7.4. Equation (7.6) localizes the numerical instability. Huge Euclidean atom coefficients can encode a moderate polynomial because they point close to null directions of G . The exact synthesis is an isometry only after coefficient space is equipped with the Gram metric.

8 Two-sided up-atom factorization of the even projection kernel

Let

$$K_N^+(x, y) = \sum_{n=0}^N \frac{P_{2n}(x)P_{2n}(y)}{h_{2n}} \quad (8.1)$$

be the projection kernel onto the even Legendre subspace of degree at most $2N$, and define

$$\tilde{K}_N^+(s, t) = \sum_{n=0}^N \frac{Q_{2n}(s)Q_{2n}(t)}{h_{2n}}. \quad (8.2)$$

Theorem 8.1 (Two-sided atomic Christoffel–Darboux factorization; new). *With $M = 4^N$,*

$$K_N^+(x, y) = \frac{1}{M^2} \sum_{k, \ell \in \mathcal{K}_M} \tilde{K}_N^+(k/M, \ell/M) u(x - k/M) u(y - \ell/M). \quad (8.3)$$

The coefficient matrix

$$C_{k\ell} = \frac{1}{M^2} \tilde{K}_N^+(k/M, \ell/M) \quad (8.4)$$

is positive semidefinite of rank $N + 1$.

Proof. Substitute the literal atomizations of $P_{2n}(x)$ and $P_{2n}(y)$ into (8.1) and interchange finite sums. In matrix form, $C = DH^{-1}D^\top$, so it is positive semidefinite. Since D has full column rank, $\text{rank } C = N + 1$. $\square$

For any $f \in L^2[-1, 1]$, the projection factors as

$$\begin{aligned} (\Pi_N^+ f)(x) &= \frac{1}{M^2} \sum_{k, \ell \in \mathcal{K}_M} u(x - k/M) \tilde{K}_N^+(k/M, \ell/M) \\ &\quad \times \int_{-1}^1 u(y - \ell/M) f(y) dy. \end{aligned} \quad (8.5)$$

Thus even Legendre projection is atom analysis, a finite positive kernel, and atom synthesis.

9 Parity anatomy of the local up-spline space

Let R denote reflection about the midpoint of the normalized cell:

$$(Rf)(t) = f(1 - t). \quad (9.1)$$

Evenness of u gives

$$R\psi_k = \psi_{1-k}. \quad (9.2)$$

Write $V_d^\pm$ and $\Pi_d^\pm$ for the ± 1 eigenspaces of R in V_d and Π_d .

Lemma 9.1 (Parity of the canonical defect; new). *The periodic residue satisfies*

$$\mathcal{E}_d(1 - t) = (-1)^{d+1} \mathcal{E}_d(t). \quad (9.3)$$

Consequently reflection acts on the one-dimensional quotient V_d/Π_d by $(-1)^{d+1}$.

Proof. In (3.9), replace t by $1 - t$ and then n by $-n$. Because n is integral and $\hat{u}$ is even, the factor is $(-1)^{d+1}$. Furthermore,

$$R\mathcal{D}_d - (-1)^{d+1}\mathcal{D}_d = (1 - t)^{d+1} - (-1)^{d+1}t^{d+1} \in \Pi_d,$$

since the leading terms cancel. This proves the quotient statement. $\square$

Theorem 9.2 (Parity-split local-space theorem; new). *For $N \geq 0$,*

$$\boxed{V_{2N}^+ = \Pi_{2N}^+, \quad V_{2N+1}^- = \Pi_{2N+1}^-} \quad (9.4)$$

The dimensions are

d	$\dim \Pi_d^+$	$\dim \Pi_d^-$	$\dim V_d^+$	$\dim V_d^-$
$2N$	$N + 1$	N	$N + 1$	$N + 1$
$2N + 1$	$N + 1$	$N + 1$	$N + 2$	$N + 1$

(9.5)

Proof. The polynomial dimensions follow by expanding in powers of $t - 1/2$. The quotient V_d/Π_d is one-dimensional by (3.10), and Lemma 9.1 specifies which parity receives the extra direction. For $d = 2N$ it is antisymmetric, so the symmetric dimensions of V_{2N} and Π_{2N} coincide; containment gives equality. For $d = 2N + 1$ it is symmetric, so the antisymmetric spaces coincide. $\square$

Corollary 9.3 (Exact polynomialization by parity projection; new). *For $f \in V_{2N}$, the symmetrization*

$$Sf = \frac{1}{2}(f + Rf) \quad (9.6)$$

belongs to Π_{2N}^+ . For $f \in V_{2N+1}$, the antisymmetrization $(f - Rf)/2$ belongs to Π_{2N+1}^- . These are the $L^2[0, 1]$ orthogonal projections onto the indicated parity sectors.

At even degree, therefore, every symmetric local up-spline is already a polynomial. This observation constructs central polynomial atoms without solving any polynomiality constraints.

10 A central reflected-pair basis and its q-determinant

10.1 Central pairs

Fix $N \geq 0$ and put

$$d = 2N, \quad M = 2^d = 4^N. \quad (10.1)$$

Define

$$S_r^{(N)}(t) = \psi_{-r}(t) + \psi_{r+1}(t), \quad r = 0, \dots, N. \quad (10.2)$$

By (9.2), each pair is symmetric about $t = 1/2$, and Theorem 9.2 gives

$$S_r^{(N)} \in \Pi_{2N}^+. \quad (10.3)$$

Only independence remains to be proved.

10.2 Endpoint-tail generating functions

The following lemma makes the endpoint reduction explicit and renders the central determinant proof self-contained once the imported annihilator theorem is accepted.

Lemma 10.1 (Endpoint-tail generating function; new). *Let $M = 2^d$, let*

$$\mathcal{A}_d(z) = \sum_{j=0}^R a_j z^j, \quad R = R_d,$$

and set

$$K = -M + 1, \quad E = K + R = M - d - 1.$$

Suppose an active coefficient vector q is supported on the initial centers $K, \dots, E-1$. Let $b_s, s = 0, \dots, d+1$, be its coordinates on the endpoint basis ψ_{E+s} . Define the reversed prefix

$$\text{rev}_p \mathcal{A}_d(z) = \sum_{j=0}^p a_{p-j} z^j. \quad (10.4)$$

Then the endpoint coordinates are the first $d+2$ coefficients of

$$\sum_{s \geq 0} b_s z^s = \frac{\sum_{p=0}^{R-1} q_{K+p} \text{rev}_p \mathcal{A}_d(z)}{\mathcal{A}_d(z)}. \quad (10.5)$$

Proof. Let n be the null coefficient sequence that agrees with q on the first R centers and is extended by the monic annihilator recurrence. After shifting K to index 0, palindromicity of $\mathcal{A}_d$ writes the recurrence as

$$\sum_{j=0}^R a_j n_{m-j} = 0, \quad m \geq R.$$

For one initial impulse at position $p < R$, let $N_p(z) = \sum_{m \geq 0} n_m z^m$. The initial equations give

$$\begin{aligned} \mathcal{A}_d(z) N_p(z) &= z^p \sum_{j=0}^{R-p-1} a_j z^j \\ &= z^p \mathcal{A}_d(z) - z^R \text{rev}_p \mathcal{A}_d(z), \end{aligned}$$

where the last step uses $a_j = a_{R-j}$. Hence

$$N_p(z) = z^p - z^R \frac{\text{rev}_p \mathcal{A}_d(z)}{\mathcal{A}_d(z)}.$$

The endpoint vector is $q - n$; since q vanishes at indices $R + s$, its coordinate there is $-n_{R+s}$. This is exactly the coefficient of z^s in $\text{rev}_p \mathcal{A}_d / \mathcal{A}_d$. Linearity proves (10.5). $\square$

10.3 The determinant certificate

Express each $S_r^{(N)}$ in the endpoint basis $\mathcal{B}_{2N}$, obtaining a $(2N+2) \times (N+1)$ coordinate matrix. Let B_N be its first $N+1$ rows.

Theorem 10.2 (Central-pair determinant; new). *For $N \geq 1$, write*

$$\mathcal{A}_{2N-1}(z) = \sum_{j \geq 0} c_j z^j, \quad C_N = 2\mathcal{A}_{2N-1}(1) = 2^{1+N(2N-1)}, \quad (10.6)$$

and define

$$\sigma_N = [z^N] \frac{1}{(1-z)\mathcal{A}_{2N-1}(z)} = [z^N] \frac{(1-z)^{2N-2}}{\prod_{m=1}^{2N-1} (1-z^{2^m})}. \quad (10.7)$$

Then

$$\boxed{\det \mathbf{B}_N = (-1)^N (C_N \sigma_N - 1)}. \quad (10.8)$$

In particular, $\det \mathbf{B}_N$ is an odd nonzero integer. At $N = 0$, the unique minor has determinant 1.

Proof. Let

$$\ell_j = \sum_{q=0}^j c_q, \quad L_N = (\ell_{s-r})_{0 \leq r \leq s \leq N},$$

with zeros above the diagonal. Since $c_0 = 1$, L_N is unit lower triangular. We show

$$L_N \mathbf{B}_N = F_N, \quad (10.9)$$

where

$$(F_N)_{sr} = \begin{cases} C_N, & r < s, \\ C_N - \ell_{r-s}, & r \geq s. \end{cases} \quad (10.10)$$

Indeed, set $\mathcal{A}_-(z) = \mathcal{A}_{2N-1}(z)$ and $H_M(z) = [M]_z = (1 - z^M)/(1 - z)$. Then

$$\mathcal{A}_{2N}(z) = \mathcal{A}_-(z) H_M(z). \quad (10.11)$$

The two impulses in the r th central pair occupy, relative to the first active center, the positions

$$p_- = M - r - 1, \quad p_+ = M + r.$$

By [Lemma 10.1](#), the generating function of the corresponding endpoint column is

$$b_r(z) = \frac{\text{rev}_{p_-} \mathcal{A}_{2N}(z) + \text{rev}_{p_+} \mathcal{A}_{2N}(z)}{\mathcal{A}_{2N}(z)}.$$

The lower Toeplitz multiplication by L_N is convolution with the first coefficients of

$$\frac{\mathcal{A}_-(z)}{1 - z}.$$

Therefore, through degree $N < M$,

$$\frac{\mathcal{A}_-(z)}{1 - z} b_r(z) = \frac{\text{rev}_{p_-} \mathcal{A}_{2N}(z) + \text{rev}_{p_+} \mathcal{A}_{2N}(z)}{1 - z^M}. \quad (10.12)$$

The denominator contributes only its constant term in this degree range. Let $S = \mathcal{A}_-(1)$ and let $a_j = [z^j] \mathcal{A}_{2N}$. Since $\deg \mathcal{A}_- = M - 2N - 1$, convolution with H_M gives

$$a_j = \sum_{i=\max(0, j-M+1)}^{\min(M-2N-1, j)} c_i.$$

For $0 \leq r, s \leq N$,

$$[z^s](\text{rev}_{p_-} \mathcal{A}_{2N} + \text{rev}_{p_+} \mathcal{A}_{2N}) = a_{M-r-1-s} + a_{M+r-s}.$$

The first term is always S . The second is S if $r < s$, and $S - \ell_{r-s}$ if $r \geq s$. Since $C_N = 2S$, this proves [\(10.10\)](#).

Because $\det L_N = 1$, it remains to evaluate $\det F_N$. For $s = 0, \dots, N - 1$, replace row s by row s minus row $s + 1$. The leading $N \times N$ block becomes the upper triangular Toeplitz matrix

$$-(c_{r-s})_{0 \leq s \leq r \leq N-1},$$

with determinant $(-1)^N$, while the last row is $(C_N, \dots, C_N, C_N - 1)$. Write

$$\frac{1}{\mathcal{A}_-(z)} = \sum_{j \geq 0} g_j z^j. \quad (10.13)$$

Inverting the triangular Toeplitz block is convolution by g_j . Its Schur complement is

$$C_N \sum_{j=0}^N g_j - 1.$$

Finally,

$$\sum_{j=0}^N g_j = [z^N] \frac{1}{(1-z)\mathcal{A}_-(z)} = \sigma_N,$$

which gives (10.8). The second formula in (10.7) follows from (2.18). Since $\mathcal{A}_-(0) = 1$, all g_j and σ_N are integers. Since C_N is even, $C_N \sigma_N - 1$ is odd and cannot vanish. $\square$

10.4 The central basis theorem

Theorem 10.3 (Central reflected-pair basis; new). *For every $N \geq 0$,*

$$\Pi_{2N}^+ = \text{span}\{S_0^{(N)}, \dots, S_N^{(N)}\}. \quad (10.14)$$

Hence, for every $0 \leq m \leq N$, there are unique rational coefficients $c_{m,r}^{(N)}$ such that, on $[-1, 1]$,

$$P_{2m}(x) = \frac{1}{M} \sum_{r=0}^N c_{m,r}^{(N)} \left[u\left(\frac{x - (2r+1)}{2M}\right) + u\left(\frac{x + (2r+1)}{2M}\right) \right], \quad M = 4^N. \quad (10.15)$$

The $N + 1$ pair blocks form a pair-count-minimal fixed basis of the $(N + 1)$ -dimensional even polynomial space.

Proof. Each $S_r^{(N)}$ lies in Π_{2N}^+ by (10.3). The nonzero minor in Theorem 10.2 proves independence. Since $\dim \Pi_{2N}^+ = N + 1$, the pairs form a basis. Their endpoint coordinates are integers and the determinant is a nonzero integer, so rational target polynomials have rational coefficients. Substitute $t = (x + 1)/2$ into $S_r^{(N)} = \psi_{-r} + \psi_{r+1}$ to obtain (10.15). Minimality follows from dimension. $\square$

For a single P_{2n} , choose $N = n$. The formula uses $n + 1$ pair blocks, or $2n + 2$ raw atoms, all at one dilation and in exact reflection pairs. This matches the raw atom count of the endpoint basis while preserving parity.

10.5 Arithmetic and the binary-partition threshold

The first values are shown in Table 2.

For $2 \leq N \leq 7$,

$$\sigma_N = (-1)^N 2^{2N-3}, \quad \det B_N = 2^{2N^2+N-2} + (-1)^{N+1}. \quad (10.16)$$

At $N = 8$, the part 8 first enters the coefficient extraction, and the simple pattern breaks. It is a finite pre-threshold phenomenon rather than a universal formula.

Table 2: Binary-partition coefficient and central-pair determinant.

N	$M = 4^N$	σ_N	$\det B_N$	simplified form
0	1	1	1	1
1	4	0	1	1
2	16	2	255	$2^8 - 1$
3	64	-8	524289	$2^{19} + 1$
4	256	32	17179869183	$2^{34} - 1$
5	1024	-128	9007199254740993	$2^{53} + 1$
6	4096	512	$2^{76} - 1$	
7	16384	-2048	$2^{103} + 1$	
8	65536	8194	$2^{134} + 2^{122} - 1$	first dyadic-part correction

10.6 Worked formulas

For $N = 0$, (10.15) is the two-atom partition

$$1 = u\left(\frac{x-1}{2}\right) + u\left(\frac{x+1}{2}\right). \quad (10.17)$$

For P_2 , take $N = 1$, $M = 4$. The normalized coefficients are $c_{1,0}^{(1)} = 848/3$ and $c_{1,1}^{(1)} = -1136/3$, so

$$P_2(x) = \frac{212}{3} \left[u\left(\frac{x-1}{8}\right) + u\left(\frac{x+1}{8}\right) \right] - \frac{284}{3} \left[u\left(\frac{x-3}{8}\right) + u\left(\frac{x+3}{8}\right) \right]. \quad (10.18)$$

The identity is exact, but its large coefficients already display severe cancellation.

For P_4 , take $N = 2$, $M = 16$. The raw amplitudes multiplying the three pairs at centers $\pm 1, \pm 3, \pm 5$ in the scaled arguments are

$$\frac{3613090304}{34425}, \quad -\frac{1818824576}{11475}, \quad \frac{1857992576}{34425}. \quad (10.19)$$

Their magnitudes are approximately 1.05×10^5 , 1.59×10^5 , and 5.40×10^4 , although $|P_4(x)| \leq 1$ on $[-1, 1]$.

11 The symmetric common-scale closed loop

Let $C^{(N)} = (c_{m,r}^{(N)})_{0 \leq m,r \leq N}$ be the change-of-basis matrix from even Legendre polynomials to central pairs, and define

$$\alpha_{N,r} = \sum_{m=0}^N u_m c_{m,r}^{(N)}. \quad (11.1)$$

Theorem 11.1 (Symmetric finite closed loop; new). *For every $N \geq 0$ and $-1 \leq x \leq 1$,*

$$S_N(x) = \frac{1}{4^N} \sum_{r=0}^N \alpha_{N,r} \left[u\left(\frac{x - (2r+1)}{2 \cdot 4^N}\right) + u\left(\frac{x + (2r+1)}{2 \cdot 4^N}\right) \right]. \quad (11.2)$$

All coefficients are rational. The formula uses $N + 1$ pair blocks and is exactly reflection symmetric at every cutoff.

Proof. Insert (10.15) into $S_N = \sum_{m \leq N} u_m P_{2m}$ and interchange the two finite sums. Rationality follows from rationality of u_m and $c_{m,r}^{(N)}$. $\square$

At $N = 1$,

$$S_1(x) = -\frac{521}{9} \left[u\left(\frac{x-1}{8}\right) + u\left(\frac{x+1}{8}\right) \right] + \frac{701}{9} \left[u\left(\frac{x-3}{8}\right) + u\left(\frac{x+3}{8}\right) \right]. \quad (11.3)$$

Corollary 11.2 (Infinite symmetric loop in grouped-cutoff form; new). *For $-1 \leq x \leq 1$,*

$$u(x) = \lim_{N \rightarrow \infty} \frac{1}{4^N} \sum_{r=0}^N \alpha_{N,r} \left[u\left(\frac{x - (2r+1)}{2 \cdot 4^N}\right) + u\left(\frac{x + (2r+1)}{2 \cdot 4^N}\right) \right]. \quad (11.4)$$

The convergence is uniform and, more strongly, occurs in every C^r norm.

The coefficients depend on the cutoff N . This is intrinsic rather than a defect: each right side is exactly the Legendre projection S_N . A cutoff-independent flattening would encounter the nonanalyticity and repeated atom obstruction described in Section 5.

12 Superalgebraic convergence and derivative loops

The repository records absolute and uniform convergence of the Legendre series. Smoothness gives convergence in the full Fréchet topology of $C^\infty[-1, 1]$.

Theorem 12.1 (Superalgebraic Legendre convergence in every derivative; new). *For every pair of nonnegative integers r, L ,*

$$\|u - S_N\|_{C^r[-1,1]} = \mathcal{O}_{r,L}(N^{-L}). \quad (12.1)$$

Consequently the symmetric atomizations (11.2) converge to u in $C^\infty[-1, 1]$.

Proof. Let $\mathcal{L} = -D(1 - x^2)D$. Repeated self-adjointness gives, for $\ell \geq 1$ and $s \geq 0$,

$$\int_{-1}^1 u(x) P_\ell(x) dx = \frac{1}{[\ell(\ell+1)]^s} \int_{-1}^1 (\mathcal{L}^s u)(x) P_\ell(x) dx. \quad (12.2)$$

All boundary terms vanish because of the factor $1 - x^2$; moreover u and all its derivatives are continuous and flat at ± 1 . Since $\|P_\ell\|_\infty \leq 1$, the coefficient in degree $2n$ satisfies

$$u_n = \mathcal{O}_s(n^{1-2s}).$$

For fixed r ,

$$\|P_\ell^{(r)}\|_\infty = P_\ell^{(r)}(1) = \frac{(\ell+r)!}{2^r r! (\ell-r)!} = \mathcal{O}_r(\ell^{2r}). \quad (12.3)$$

Therefore

$$\sum_{n>N} |u_n| \|P_{2n}^{(r)}\|_\infty = \mathcal{O}_{r,s} \left(\sum_{n>N} n^{1-2s+2r} \right).$$

Choosing s arbitrarily large proves (12.1). Each central atomization equals S_N identically, so the same estimate applies. $\square$

Corollary 12.2 (Derivative form of the literal closed loop; new). *For every fixed $r \geq 0$,*

$$u^{(r)}(x) = \sum_{n=0}^{\infty} \frac{u_n}{4^n} \sum_{|k| < 2 \cdot 4^n} Q_{2n}(k/4^n) u^{(r)}(x - k/4^n), \quad (12.4)$$

with uniform convergence of the grouped blocks on $[-1, 1]$.

Proof. Differentiate the finite identity (4.7) and use Theorem 12.1 on the resulting Legendre derivative series. $\square$

The central version includes the scale derivative factor. With $M = 4^N$,

$$S_N^{(r)}(x) = \frac{1}{M(2M)^r} \sum_{j=0}^N \alpha_{N,j} \left[u^{(r)}\left(\frac{x - (2j+1)}{2M}\right) + u^{(r)}\left(\frac{x + (2j+1)}{2M}\right) \right], \quad (12.5)$$

and these expressions converge uniformly to $u^{(r)}$.

13 Exact and numerical experiments

The accompanying program `legendre_up_experiments.py` separates exact symbolic arithmetic from floating diagnostics. Its exact layer uses Python integers and SymPy rationals; the plots use the cosine expansion

$$u(x) = \frac{1}{2} + \sum_{k \geq 0} \hat{u}\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi x), \quad |x| \leq 1, \quad (13.1)$$

with Fourier coefficients evaluated from (2.1).

13.1 Determinant and reconstruction certificates

Direct endpoint-recurrence calculations agree with (10.8) as shown in Table 3.

Table 3: Exact direct verification of the central determinant.

N	M	rank	direct determinant	formula
0	1	1	1	1
1	4	2	1	1
2	16	3	255	255
3	64	4	524289	524289
4	256	5	17179869183	17179869183
5	1024	6	9007199254740993	9007199254740993

The code also verifies every unused endpoint row after solving for the central Legendre coefficients. Thus the low-degree reconstructions are checked as identities in the exact local quotient, not merely at sample points.

The normalized correction

$$R_N = \frac{\sigma_N}{(-1)^N 2^{2N-3}}, \quad N \geq 2,$$

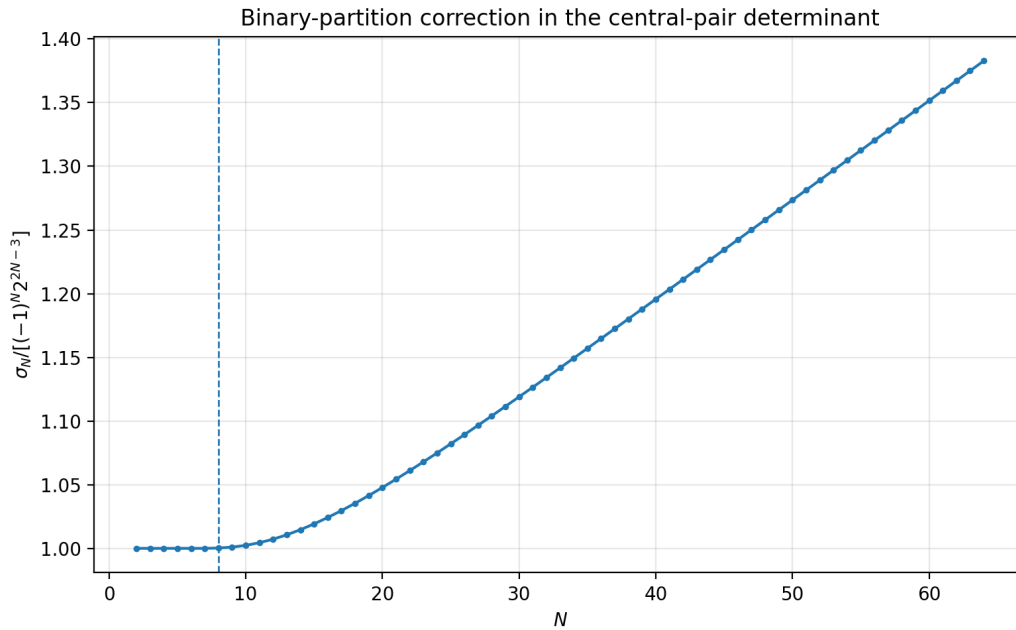


Figure 1: The normalized binary-partition correction R_N through $N = 64$. The dashed line marks the first new dyadic part at $N = 8$.

is exactly 1 through $N = 7$ and first changes at $N = 8$.

Exact coefficient extraction verifies the alternating sign $(-1)^N \sigma_N > 0$ for every $2 \leq N \leq 128$. This remains evidence rather than proof for all N .

13.2 Cancellation in exact central atomizations

The central formulas are exact, but double precision rapidly loses digits as the signed amplitudes grow.

Table 4: Double-precision evaluation of exact central formulas for P_{2N} .

N	largest raw pair amplitude	maximum reconstruction error
1	9.47×10^1	9.22×10^{-14}
2	1.59×10^5	2.38×10^{-10}
3	2.94×10^{10}	4.21×10^{-5}
4	1.80×10^{17}	2.60×10^2

The Gram identity (7.6) explains the phenomenon. Stable algorithms should preserve the Legendre blocks, use exact or multiprecision arithmetic, or orthogonalize the atom coefficients in the G metric before evaluation.

13.3 Legendre convergence

The plotted curves use the polynomial side rather than raw central atoms. Mathematically they are the same functions; numerically they inhabit very different coordinate systems.

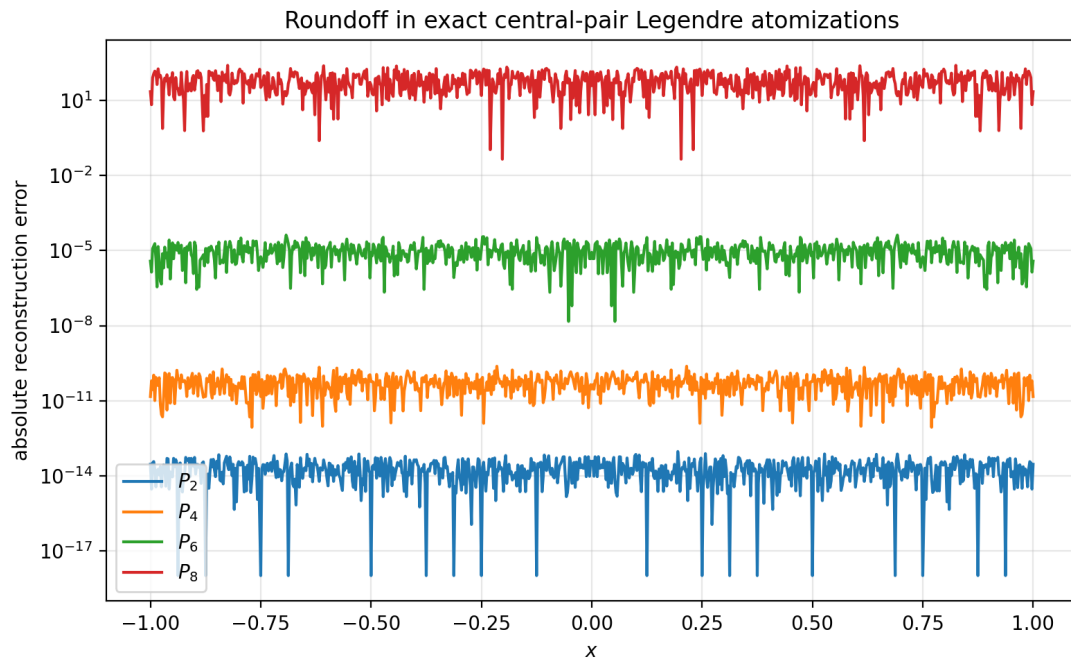


Figure 2: Pointwise floating residuals for exact central-pair representations. The deterioration is cancellation in near-null Gram directions, not failure of the identities.

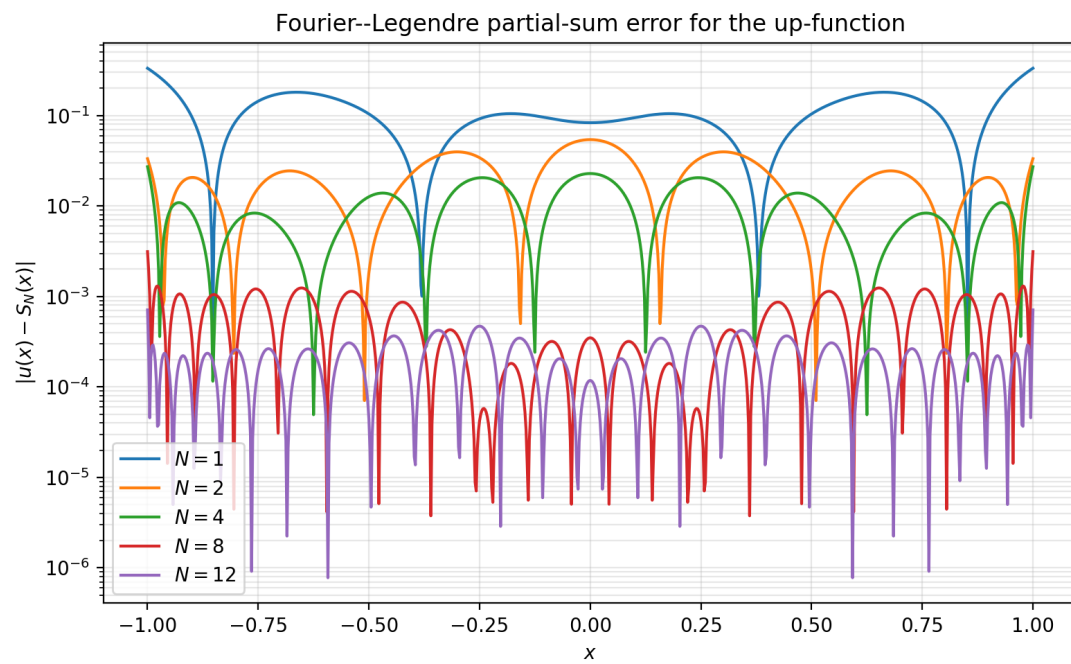


Figure 3: Error of the stable polynomial partial sum S_N . The convergence is superalgebraic but not necessarily monotone at each point or cutoff.

13.4 An exact spectral transition at degree twelve

The Q_n are not a scalar orthogonal family, so real-rootedness is not protected by Favard theory. Exact Sturm counts in rational arithmetic give the first transition.

Proposition 13.1 (First nonreal zeros; exact computer-assisted certificate). *For $1 \leq n \leq 11$, Q_n has exactly n real roots. The polynomial Q_{12} has exactly eight real roots and therefore four nonreal roots.*

Proof. The accompanying script constructs each Q_n over $\mathbb{Q}$ and applies Sturm’s theorem. For Q_{12} , the exact Sturm chain has degrees $12, 11, \dots, 0$ and leading-coefficient signs

$$+, +, +, +, +, +, +, +, -, +, +, +, +.$$

At $+\infty$ this sequence has two sign variations. At $-\infty$, parity of the degrees gives

$$+, -, +, -, +, -, +, -, -, -, +, -, +,$$

with ten variations. Sturm’s theorem therefore gives $10 - 2 = 8$ real roots. The same exact calculation gives n real roots for every $n \leq 11$. The file `Q12_sturm_certificate.txt` records the certificate and `sturm_real_root_counts.csv` records the complete table through $n = 20$. $\square$

High-precision numerical roots locate the first conjugate quartet near

$$\pm 1.58181382985647 \pm 0.189564695738147 i. \quad (13.2)$$

Exact nonreal-root counts for $n = 12, \dots, 20$ are

$$4, 4, 4, 8, 8, 8, 8, 12, 12. \quad (13.3)$$

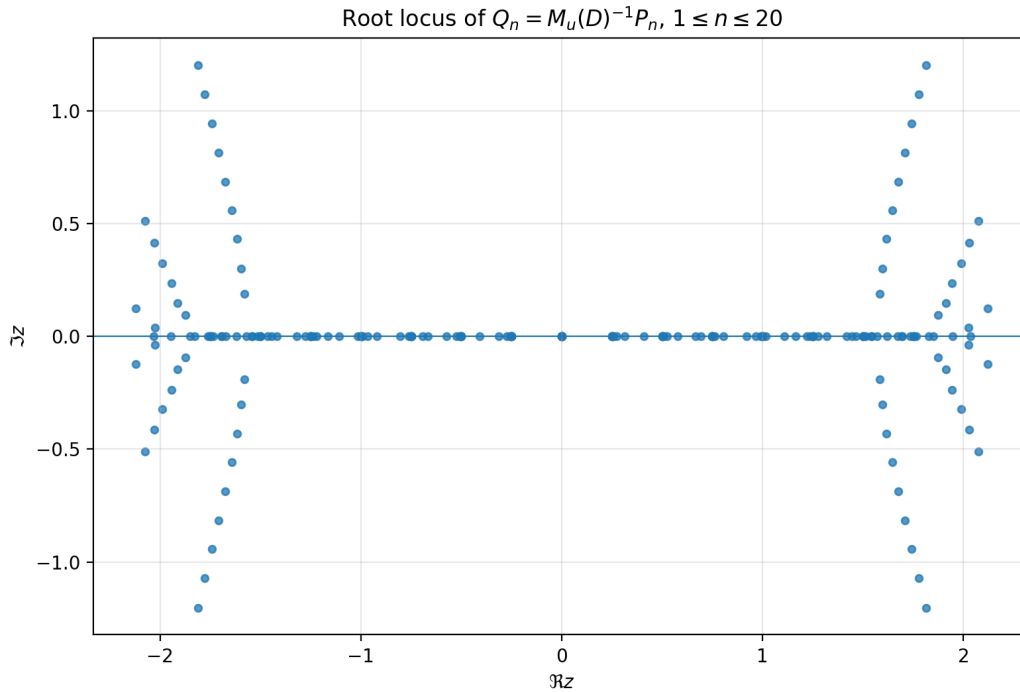


Figure 4: Numerical root locus of $Q_n = M_u(D)^{-1}P_n$ for $1 \leq n \leq 20$. Real coefficients and parity force nonreal zeros into quartets.

This transition is consistent with the operator interpretation: Q_n is orthogonal in the pullback coordinate X_u , not under scalar multiplication by x .

14 Minimal central Gram geometry and structural interpretation

The literal lattice is ideal for explicit sampling identities, while the central basis is dimension-optimal for the even finite-dimensional space. It therefore has its own exact Gram and kernel calculus.

Define the physical central pair blocks

$$\mathcal{S}_r^{(N)}(x) = \frac{1}{M} \left[u \left(\frac{x - (2r + 1)}{2M} \right) + u \left(\frac{x + (2r + 1)}{2M} \right) \right], \quad M = 4^N, \quad (14.1)$$

and let $\mathbf{S}_N(x)$ be the column vector of these $N + 1$ functions. Write

$$\mathbf{P}_N(x) := (P_0(x), P_2(x), \dots, P_{2N}(x))^T, \quad C^{(N)} := (c_{m,r}^{(N)}).$$

Then

$$\mathbf{P}_N(x) = C^{(N)} \mathbf{S}_N(x). \quad (14.2)$$

Set

$$\mathcal{G}_N = \int_{-1}^1 \mathbf{S}_N(x) \mathbf{S}_N(x)^T dx, \quad H_N^+ = \text{diag}(h_0, h_2, \dots, h_{2N}). \quad (14.3)$$

Theorem 14.1 (Central Gram factorization and dual basis; new). *One has*

$$\boxed{C^{(N)} \mathcal{G}_N C^{(N)T} = H_N^+, \quad \mathcal{G}_N^{-1} = C^{(N)T} (H_N^+)^{-1} C^{(N)}}. \quad (14.4)$$

The L^2 -dual functions to the central pairs are

$$\eta_r^{(N)}(x) = \sum_{m=0}^N \frac{c_{m,r}^{(N)}}{h_{2m}} P_{2m}(x), \quad (14.5)$$

and satisfy

$$\int_{-1}^1 \eta_r^{(N)}(x) \mathcal{S}_s^{(N)}(x) dx = \delta_{rs}. \quad (14.6)$$

Finally, the even projection kernel has the pair-minimal factorization

$$\boxed{K_N^+(x, y) = \mathbf{S}_N(x)^T \mathcal{G}_N^{-1} \mathbf{S}_N(y)}. \quad (14.7)$$

Proof. Integrate (14.2) against its transpose and use Legendre orthogonality. Since $C^{(N)}$ is invertible, the inverse identity follows. In vector form, the proposed dual family is $C^{(N)T} (H_N^+)^{-1} \mathbf{P}_N = \mathcal{G}_N^{-1} \mathbf{S}_N$; integrating against $\mathbf{S}_N$ gives the identity matrix. Finally,

$$K_N^+(x, y) = \mathbf{P}_N(x)^T (H_N^+)^{-1} \mathbf{P}_N(y)$$

and substitution of (14.2) gives (14.7). □

This theorem compresses the two-sided factorization (8.3) from $4M - 1$ literal shifts in each variable to $N + 1$ reflection-pair blocks in each variable. The price is that $\mathcal{G}_N$ becomes severely ill-conditioned.

14.1 Three exact coordinate systems

At fixed cutoff N , the same polynomial S_N has three exact descriptions:

1. the orthogonal Legendre coordinates $u_0, \dots, u_N$, stable in L^2 and spectrally natural;
2. the literal shift lattice, whose coefficients are sampled values of $C_N = \mathcal{D}_u S_N$ and whose redundancy exposes Poisson/sampling structure;

3. the central-pair coordinates $\alpha_N = C^{(N)\top} u$, which are pair-count minimal and parity adapted but cancellation dominated.

The coordinate transforms are exact; stability is not invariant under a change of coordinates. In particular, a huge determinant does not imply a well-conditioned basis when the columns themselves have enormous norms and nearly cancel.

14.2 How the q-product enters

The determinant mechanism has a clean hierarchy:

$$\text{dyadic zeros of } \hat{u} \implies \mathcal{A}_d(z) = \prod_{m \leq d} [2^m]_z \implies \mathcal{A}_d^{-1} \implies \sigma_N \implies \det B_N.$$

The Thue–Morse product controls the numerator side of dyadic cancellation, while reciprocal q -integer products count restricted binary partitions. The central basis therefore turns an analytic compact-support problem into an integer coefficient-extraction problem.

14.3 Why closure does not force analyticity

Both (5.4) and (11.4) express u in terms of copies of itself, but neither is a finite-scale analytic functional equation with uniformly summable scalar coefficients. The literal loop converges only after finite cancellation blocks are retained; the central loop changes scale and coefficients with N . These features are compatible with the classical fact that the Fabius/up function is smooth but nowhere analytic in the relevant sense.

15 Conjectures and frontier problems

The following statements are motivated by the exact formulas and computations, but are not proved here.

Conjecture 15.1 (Alternating sign of the central coefficient). *For every $N \geq 2$,*

$$(-1)^N \sigma_N > 0. \tag{15.1}$$

Exact computation verifies this for all $2 \leq N \leq 128$. A proof may require a sign-reversing involution for

$$[z^N](1-z)^{2N-2} \prod_{m \geq 1} (1-z^{2^m})^{-1}$$

or a contour argument adapted to the dyadic singularity set.

Conjecture 15.2 (Subexponential binary-partition correction). *Let*

$$R_N = \frac{(-1)^N \sigma_N}{2^{2N-3}}.$$

Then $R_N \rightarrow \infty$ but

$$\log R_N = o(N). \tag{15.2}$$

Moreover, after subtraction of a smooth main term in $\log_2 N$, the remainder has a bounded log-periodic fluctuation characteristic of Mahler-type binary partition products.

The data through $N = 128$ show positivity and slow growth, but are far too short to distinguish competing asymptotic scales. Mellin analysis of the reciprocal q -product is a promising route.

Conjecture 15.3 (Central Gram conditioning). *For the pair-minimal Gram matrix $\mathcal{G}_N$,*

$$\log \kappa_2(\mathcal{G}_N) = \Theta(N^2). \quad (15.3)$$

Equivalently, pair-minimal central synthesis should have condition number $\exp(\Theta(N^2))$.

This is consistent with the $2^{\Theta(N^2)}$ arithmetic scales in C_N and the observed coefficient explosion, but determinant size alone is not enough to prove it. A proof should combine singular-value bounds for the endpoint recurrence with the exact Gram factorization (14.4).

Conjecture 15.4 (Unbounded complex-zero population). *The number of nonreal zeros of Q_n tends to infinity with n , and every Q_n has only simple zeros.*

The first nonreal quartet occurs exactly at degree 12, and the exact counts through degree 20 are given in (13.3). A natural program is to study the eigenvalue problem for scalar multiplication x in the pullback-orthogonal basis, viewed as a nonnormal perturbation of the self-adjoint X_u recurrence.

Problem 15.5 (Transmuted Christoffel asymptotics). *Determine the large- d behavior of*

$$\tilde{K}_d(x, x) = \sum_{n=0}^d \frac{Q_n(x)^2}{h_n}$$

in the bulk, at $x = 0$, and near ± 1 . The factorial growth of deconvolution suggests a scale very different from the classical Legendre Christoffel function. The operator identity (6.20) should replace the usual scalar quotient formula.

Problem 15.6 (Optimal symmetric pair placement). *Among all $N + 1$ reflection pairs $\psi_k + \psi_{1-k}$ selected from the active one-cell dictionary at degree $2N$, determine which choices minimize κ_2 of the L^2 Gram matrix, the endpoint-coordinate matrix, or the L^∞ synthesis operator. The central contiguous choice is arithmetically canonical, but need not be numerically optimal.*

Problem 15.7 (Stable evaluation and precision budgets). *Construct an algorithm that evaluates the central loops without expanding near-null directions in ordinary Euclidean coordinates. Candidates include G -orthogonal QR, exact modular reconstruction, compensated pair summation, and direct use of the dual basis (14.5). Balancing spectral truncation error against coefficient amplification leads to implicit cutoff relations of the type already treated by Lambert–W asymptotics elsewhere in the repository.*

Problem 15.8 (Jacobi, Gegenbauer, and other symmetric bases). *For any symmetric polynomial family R_n with a differential or bispectral operator, define $\tilde{R}_n = \mathcal{D}_u R_n$. The finite up-synthesis and pullback orthogonality extend immediately, while conjugation transports the spectral operator. Work out explicit Jacobi/Gegenbauer analogues of (6.20), determine their scalar-Favard obstructions, and compare their central-pair conditioning.*

Problem 15.9 (Beyond the dyadic atom). *For atomic functions generated by bases $a \geq 3$ or by nonuniform product scales, identify the analogues of $\mathcal{A}_d$, the parity split, and the central determinant. One expects reciprocal products related to restricted a -ary partitions and generalized Thue–Morse/Prouhet sequences. The dyadic case suggests that determinant arithmetic may classify when symmetry-adapted minimal dictionaries exist.*

Problem 15.10 (Formal verification). *The central determinant proof is well suited to formalization: it reduces to a palindromic recurrence, a truncated generating-function identity, a Toeplitz row reduction, and integer parity. A Lean development could import the repository’s existing Fabius/up infrastructure and certify the new basis, determinant, Gram, and kernel identities independently of numerical evidence.*

16 Conclusions

The requested loop closes in its literal form by two exact transforms:

$$P_{2n} \xrightarrow{\mathcal{D}_u \text{ and sampling}} \text{finite shifted up-atoms}, \quad u \xrightarrow{\text{Legendre analysis}} (u_n)_{n \geq 0}.$$

Their composition is the blockwise self-representation (5.4). The main analytical warning is that closure lives in the grouped Legendre topology, not in an unconditional scalar atom series.

The new results sharpen this picture in three directions. First, the polynomials $Q_n = \mathcal{D}_u P_n$ are ordinary Legendre polynomials viewed through a nonlocal but positive pullback geometry. Their true coordinate operator is X_u , their spectral operator is $\mathcal{L}_u$, and their Christoffel–Darboux identity is operator valued. The exact Favard obstruction and the first nonreal zeros at degree 12 show concretely why no scalar-weight theory can replace this geometry.

Second, literal atom synthesis has an exact finite matrix calculus: $D^\top G D = H$, $BD = I$, and the even projection kernel has a positive rank- $(N + 1)$ two-sided up-atom factorization. These identities separate functional exactness from Euclidean coefficient conditioning.

Third, reflection removes the one-dimensional nonpolynomial defect at every even degree. This creates a pair-minimal central basis, whose independence is controlled by the odd integer determinant $(-1)^N (C_N \sigma_N - 1)$. The determinant exposes a new bridge from the Rvachev atom to reciprocal q -integer products and binary partitions. It also yields a symmetric common-scale loop for every Legendre partial sum, converging superalgebraically in all derivatives.

The resulting picture is not merely a loop but a commuting network among orthogonal polynomials, deconvolution Appell calculus, dyadic sampling, q -products, parity quotients, and finite Gram geometry. The most promising next steps are the sign/asymptotic analysis of σ_N , stable central coordinates, complex-zero asymptotics for Q_n , and formal verification.

A Exact coefficient algorithms

This appendix records the recurrences used by the accompanying program.

A.1 Moments and reciprocal-MGF coefficients

Given cumulants κ_r , the complete Bell recurrences are

$$\mu_0 = 1, \quad \mu_n = \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \mu_{n-r}, \quad (\text{A.1})$$

$$\gamma_0 = 1, \quad \gamma_n = - \sum_{r=1}^n \binom{n-1}{r-1} \kappa_r \gamma_{n-r}. \quad (\text{A.2})$$

These generate M_u and $1/M_u$ in exact rational arithmetic.

A.2 Deconvolved Legendre coefficients

For a polynomial p of degree d ,

$$\mathcal{D}_u p = \sum_{j=0}^d \gamma_j \frac{p^{(j)}}{j!}. \quad (\text{A.3})$$

Equation (4.3) is the specialization to P_{2n} .

A.3 Endpoint reduction

Write $\mathcal{A}_d(z) = \sum_{j=0}^R a_j z^j$. Given the first R active coefficients $n_0, \dots, n_{R-1}$, extend by

$$n_m = - \sum_{j=1}^R a_j n_{m-j}, \quad m \geq R, \quad (\text{A.4})$$

using palindromicity to match the repository’s shift convention. Subtract the resulting null sequence from the active vector; the final $d + 2$ entries are the endpoint coordinates. The generating form is (10.5).

A.4 Binary-partition coefficient

Let p_j be the coefficients of

$$\prod_{m \geq 1} (1 - z^{2^m})^{-1} = \sum_{j \geq 0} p_j z^j. \quad (\text{A.5})$$

They are computed by the unbounded-coin recurrence for parts 2, 4, 8, $\dots$. Then

$$\sigma_N = \sum_{j=0}^N (-1)^j \binom{2N-2}{j} p_{N-j}. \quad (\text{A.6})$$

Only powers of two not exceeding N are required.

A.5 Central coefficients

For fixed N , form the endpoint-coordinate matrix of the pairs $S_r^{(N)}$. Solve the nonsingular $(N + 1) \times (N + 1)$ minor for each target $P_{2m}(2t - 1)$. The program then verifies all remaining endpoint rows exactly. The closed-loop vector is

$$\alpha_N = C^{(N)\top} (u_0, \dots, u_N)^\top. \quad (\text{A.7})$$

B Reproducibility inventory

The archive accompanying this report contains:

`legendre_rvachev_closed_loop.tex`

The complete LaTeX source.

`legendre_rvachev_closed_loop.pdf`

The rendered report.

`legendre_up_experiments.py`

Fully commented exact and numerical experiment code.

`numerical_results.txt`

Human-readable exact values and diagnostics.

CSV tables

Exact central coefficients, closed-loop coefficients, determinant checks, σ_N data, Legendre coefficients, Sturm counts, roots, and floating errors.

PNG figures

The determinant correction, roundoff, Legendre convergence, and root locus figures used in the report.

Q12_sturm_certificate.txt

The exact degree/sign certificate used in [Proposition 13.1](#).

The calculations can be regenerated with

```
python legendre_up_experiments.py
```

The report can then be compiled with a standard modern LaTeX installation. All exact claims in the numerical section are independent of Fourier truncation; only the plotted function values use floating arithmetic.

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